



S&P GLOBAL AND CRISIL CAMPUS HACKATHON

RiskSignal Engine

Explainable financial event signals connected to a
reproducible synthetic portfolio stress workflow

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Version 0.8.0

Problem and implemented approach

Monitoring gap

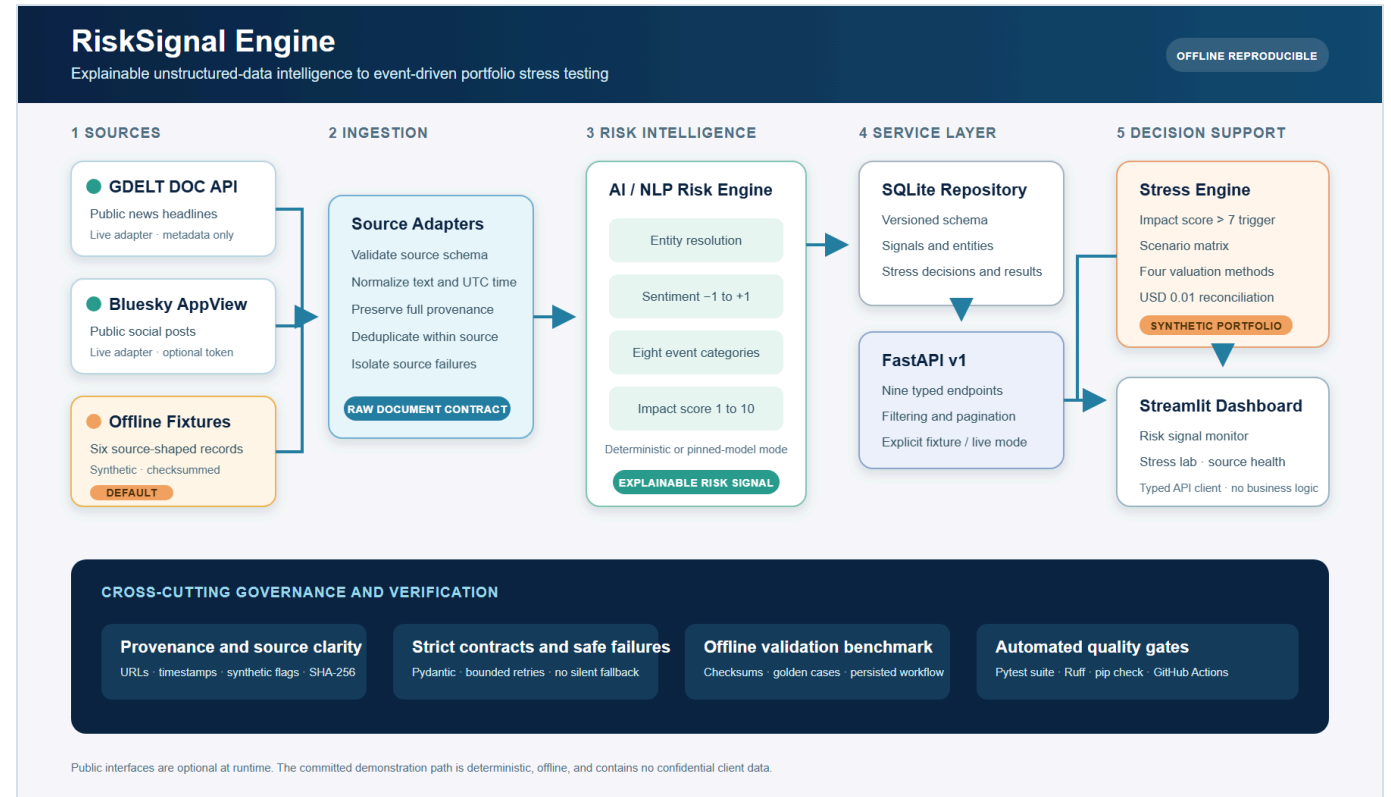
News and social text need consistent event, entity, sentiment, and provenance records before an analyst can act.

Risk signal

The engine produces an explainable 1 to 10 impact score and retains every weighted factor.

Downstream use

Scores above 7 select a configured scenario and stress the matching fictional portfolio scope.



Data and mode boundaries

Public live interfaces

GDELT news and Bluesky public posts remain optional and externally dependent.

Committed review path

Six baseline records and four separate replay records carry synthetic provenance and checksums.

Explicit execution

Fixture, replay, live, deterministic, and model modes never substitute for one another.

Dataset claim

No proprietary client data appears in the repository or presentation.

Connection

API 0.8.0 connected

Mode: offline fixtures

Refresh dashboard

Signal filters

Event type

All

Minimum impact

Source

All

Entity

All

Filters apply to every signal view and the Stress Lab selector.

RiskSignal Monitor

Explainable financial-risk signals and event-driven portfolio stress testing

Risk signals Stress lab Source health

Healthy sources2 / 2

Latest querybanking stress

Last completed (UTC)2026-10-04 06:21

Source	Type	Status	Fetched	Normalized	Rejected	Duplicates	Completed (UTC)
replay-news	news	Healthy	2	2	0	0	2026-10-04T06:21:56.606473+00:00
replay-social	social	Healthy	2	2	0	0	2026-10-04T06:21:56.607119+00:00

Analysis mode

NLP mode

deterministic

Reproducible fixture refresh

This action explicitly uses committed fixtures with `deterministic` NLP. It never falls back between source or NLP modes.

Ingestion query

portfolio risk

Ingest and analyze fixtures

Synthetic banking-stress replay

This separate offline mode runs four project-authored records inspired by the general March 2023 banking-stress pattern. It is hypothetical, synthetic, and not copied news.

Recorded evidence

Evidence	Recorded result	Interpretation
Deterministic benchmark	Accuracy 0.6767 Macro-F1 0.4302	Transparent baseline on 2,264 expert labelled sentences
Pinned FinBERT benchmark	Accuracy 0.9717 Macro-F1 0.9625	Training overlap means this is not an out of sample estimate
Synthetic NLP regression	8 of 8 exact matches	All event categories covered for event, sentiment, and entity outputs
Offline validator	9 of 9 artifacts USD 0.00 reconciliation	Reproducibility and contract evidence with no network in tests
API and dashboard	10 endpoints 3 workspaces	Persisted signals, stress decisions, and source health remain inspectable

Both benchmark modes produced 0 of 2,264 scores above 7. Financial PhraseBank supplies sentiment labels, not ground truth trigger labels.

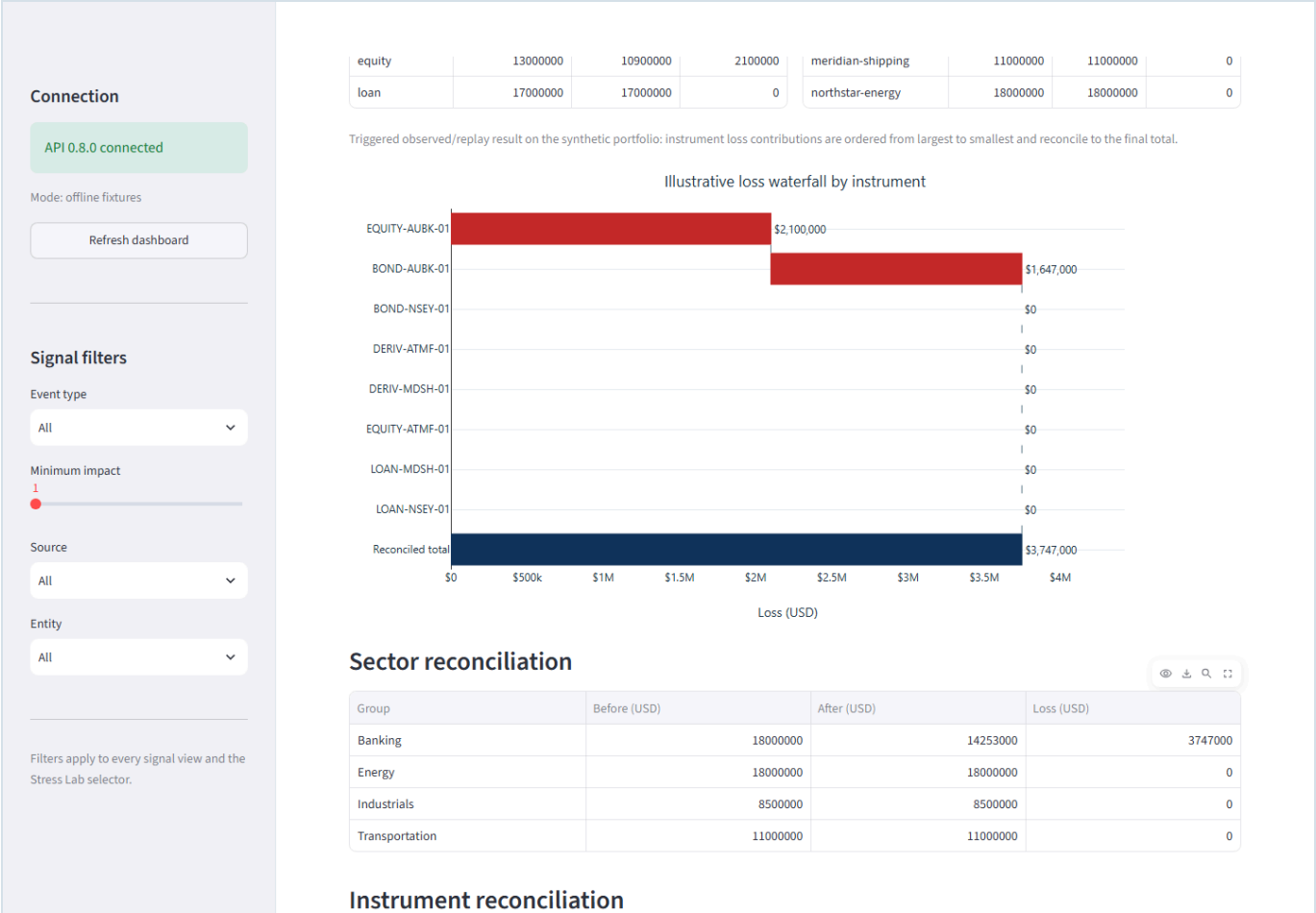
Organic replay stress result

Four fictional banking stress records produce four impact-9 signals through the unchanged deterministic engine.

USD 3.747M

Illustrative Aurora Bank portfolio loss with a USD 0.00 reconciliation difference.

The positions, issuers, replay text, and scenario shocks are synthetic. The result demonstrates implementation behavior, not a forecast.



Source: docs/results.md and committed dashboard evidence

5

Reviewer path and limits

The reviewer demo is one command.

```
python -m risk_engine demo
```

Impact direction: absolute sentiment is direction agnostic. Positive language can raise impact magnitude without being harmful.

Valuation: derivative P&L uses signed delta exposure and positive DV01 as USD loss per +1 bp rate rise.

Scope: the prototype has no investment advice, regulatory calibration, client portfolio, or production access controls.

The screenshot displays the 'RiskSignal Monitor' web application. On the left is a sidebar with a 'Connection' section showing 'API 0.8.0 connected' and a 'Refresh dashboard' button. Below this is the 'Signal filters' section with dropdowns for 'Event type' (set to 'All'), 'Source' (set to 'All'), and 'Entity' (set to 'All'). A 'Minimum impact' slider is also present. The main content area is titled 'RiskSignal Monitor' and includes a subtitle 'Explainable financial-risk signals and event-driven portfolio stress testing'. It has tabs for 'Risk signals', 'Stress lab' (which is active), and 'Source health'. A yellow warning box states: 'Synthetic scenario: shocks and portfolio positions are project-authored assumptions for demonstration. They are not calibrated regulatory scenarios.' Below this is the 'Hypothetical what-if' section, which includes a disclaimer: 'Hypothetical simulation only: these user-selected assumptions are not an observed or replay signal, and neither the assumptions nor result are persisted.' It features a 'Hypothetical event' dropdown set to 'Geopolitical', a 'Hypothetical entities' dropdown set to 'atlas-manufacturing', and a 'Hypothetical impact score' slider set to 8. A red 'Run hypothetical simulation' button is located below the slider. A note at the bottom of this section says: 'Run the controls above to calculate a non-persisted hypothetical result.' The final section is 'Observed or replay signal stress', which includes a 'Trigger signal' dropdown.